In the following we refer to all absorbed radiation energy as the radiation dose independent of whether the incident radiation is charged or uncharged particles or photons.

Radiation chemical yield is described in terms of *G-values*. Originally $G(x)$ was defined as the number of molecules of x transformed per 100 eV absorbed energy (a practical notation as most reactions have G-values of <10). In the SI system the symbol $G(x)$ is the same but the unit is mol/J. The conversion factor between the two units is 1 mol/J $= 9.649 \times 10^6$ molecules per 100 eV absorbed. Today the symbol $g(x)$ usually refer to the number per 100 eV and $G(X)$ to mol/J.

8.4. METALS

Metals consist of a solid lattice of atoms whose valence electrons cannot be considered to belong to any particular atom, but rather to a partially filled energy band (the conduction band) established by the total lattice network. Interaction of radiation with the metal can cause *excitation* of bound electrons in the atoms to the conduction band. This, however, due to the excellent conducting properties of metals do not lead to prevailing electron deficiencies resulting in e.g. colour changes.

While irradiation by γ-rays and electrons has little influence on metallic properties, heavy particles cause serious damage through their collision with atoms in the metal lattice network. This results in *displacements* of the atoms from their lattice positions. The number of displacements (n_{disp}) depends on the amount of energy transferred in the collision event (E_{tr}) to the recoiling (target) atom, and the energy required for moving this atom from its lattice position. This so-called *displacement energy* averaged over all directions in the crystal lattice (E_{disp}) is 10–30 eV for most metallic materials. According to the Kinchin-Pease rule,

$$n_{\text{disp}} \leq \tfrac{1}{2} E_{\text{tr}} / E_{\text{disp}} \tag{8.14}$$

The maximum energy transferred can be calculated assuming purely elastic collisions between hard spheres (§10.1). Thus for a 1.5 MeV fission neutron, E_{tr}(max) is 425 keV in C, 104 keV in Fe, and 25 keV in U. With $E_{\text{disp}} \approx 25$ eV, up to 8500, 2080 and 500 displacements, respectively, occur in these metals due to the absorption of a fission neutron (neglecting nuclear reactions). In practice the numbers are somewhat smaller, especially at the higher E_{tr} (where it may be about one third of the calculated value). The reason for this is the recombination of defects at the end of the collision cascade. Atomic displacements cause many changes in the properties of metals. Usually electrical resistance, volume, hardness and tensile strength increase, while density and ductility decrease.

The microcrystalline properties of metals are particularly influenced by irradiation. Although low-alloy steel in modern reactor tanks are rather radiation resistant (provided

they are free of Cu, P, Ni, Mn and S impurities). Significant radiation damage in the reactor vessels due to impurities of Ni and Mn has been observed in both BWR and VVER reactors. Stainless steel (e.g. of the 18% Cr, 8% Ni type) has been found to become brittle upon irradiation due to the formation of microscopic helium bubbles, probably due to n,α reactions in ^{54}Fe and impurities of light elements (N, B, etc.). This behaviour is accentuated for metallic uranium in reactors because of the formation of fission products, some of which are gases. As a result of this radiation effect it is not possible to use uranium metal in modern power reactors, where high radiation doses are accumulated in a very short time. The fuel elements for power reactors are therefore made of non-metallic uranium compounds. However, in some early fast reactors, e.g. EBR II, metallic uranium fuel was used successfully by increasing the gap between the fuel and the can and filling it with sodium.

The displaced atoms may return to their original lattice positions through diffusion if they are not trapped in energy wells requiring some activation energy for release. Such energy can be provided by heating or by irradiation with electrons or γ-rays (these do not cause new displacements). This "healing" of particle radiation damage is commonly referred to as annealing. The *thermal annealing rate* increases with temperature as does *radiation annealing* with radiation dose. Doses in the 10 kGy range are usually required for appreciable effect.

8.5. INORGANIC NON-METALLIC COMPOUNDS

The time for a high energy particle to pass by an atom is $\leq 10^{-16}$ s. In this time the atom may become excited and/or ionized, but it does not change position (the *Franck-Condon principle*) provided there is no direct collision. The excited atoms are de-excited through the emission of *fluorescence radiation*, usually within 10^{-8} s. The ionization can result in simple trapping of the electrons and production of "electron holes" in the lattice, especially at impurity sites. The local excess (or deficiency) of charge produced in this way leads to electronic states with absorption bands in the visible and ultraviolet regions of the spectrum. For example, irradiation of LiCl results in a change of the colour of the compound from white to yellow. Similarly, LiF becomes black, KCl blue, etc. The irradiation of ionic crystals also leads to changes in other physical properties much as conductivity, hardness, strength, etc. Frequently, heating (sometimes room temperature is enough) returns the properties and colour to the normal state (or close to it) accompanied by the emission of light; this forms the basis for a radiation dose measurement technique named "thermoluminescence dosimetry" (§8.10).

Following a collision between a heavy particle (n, p, etc.) and an absorber atom in a crystalline material the recoiling ion produces lattice vacancies and, upon stopping, may occupy a non-equilibrium interstitial position (Fig. 8.3). The localized dissipation of energy can result in lattice oscillations, terminating in some reorientation of the local